

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE**Supplementary Winter Examination – 2025****Course: BTECH****Branch : COMP/CSE****Semester : VI****Subject Code & Name: BTCOC602 Computer Networks****Max Marks: 60****Date: 12/01/2026****Duration: 3 Hr.****Instructions to the Students:**

1. Each question carries 12 marks.
2. Question No. 1 will be compulsory and include objective-type questions.
3. Candidates are required to attempt any four questions from Question No. 2 to Question No. 6.
4. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
5. Use of non-programmable scientific calculators is allowed.
6. Assume suitable data wherever necessary and mention it clearly.

		(Level/CO)	Marks
Q. 1	Objective type questions. (Compulsory Question)		12
1	A computer network is mainly used for: a. Data storage b. Resource sharing c. Program execution d. Hardware manufacturing		1
2	Connection-oriented service is best represented by: a. UDP b. IP c. TCP d. ICMP		1
3	TCP/IP model consists of: a. 7 layers b. 5 layers c. 4 layers d. 3 layers		1
4	Wi-Fi standard belongs to: a. 802.3 b. 802.5 c. 802.11 d. 802.16		1
5	Bluetooth standard is: a. 802.11 b. 802.15.1 c. 802.16 d. 802.3		1
6	Frame Relay operates at: a. Physical layer b. Data link layer c. Network layer d. Transport layer		1
7	Flow control ensures: a. Error-free transmission b. Sender does not overwhelm receiver c. Faster transmission d. Security		1
8	Routers operate mainly at:		1

	a. Data link layer	b. Network layer	c.Transport layer	d. Application layer		
9	UDP is:					1
	a. Reliable	b. Connection-oriented	c. Connectionless	d. Secure		
10	POP protocol is mainly for:					1
	a. Sending mail	b. Receiving mail	c. File upload	d. Web browsing		
11	Public key cryptography uses:					1
	a. One shared key	b. Two keys	c. No key	d. Password only		
12	Firewalls are used to:					1
	a. Increase bandwidth	b. Control network access	c. Detect errors	d. Speed transmission		
Q. 2	Solve the following.					12
A)	Explain applications of computer networks in detail.					6
B)	Explain TCP/IP reference model and compare it with OSI model.					6
Q.3	Solve the following.					12
A)	Explain Ethernet architecture and working.					6
B)	Explain wireless LAN technologies: Wi-Fi and Wi-MAX.					6
Q. 4	Solve Any Two of the following.					12
A)	Explain Data Link Layer design issues in detail.					6
B)	Explain framing, flow control and error control.					6
C)	Compare error detecting codes and error correcting codes.					6
Q.5	Solve Any Two of the following.					12
A)	Explain routing algorithms: distance vector and link state.					6
B)	Explain TCP and UDP with comparison.					6
C)	Explain Quality of Service (QoS) and its components.					6

Q. 6	Solve Any Two of the following.		12
A)	Explain DNS and HTTP in detail.		6
B)	Explain network security concepts and authentication.		6
C)	Explain digital signatures, certificates and firewalls in detail.		6
	*** End ***		